

Project 2: Velocity Estimates

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Background:

The purpose of this project is to examine the 2 different methods of estimating the velocity of the end of the cantilever beam using acceleration and position measurements. By building and understanding both the accelerator and the strain gauge circuits, we can deepen our understanding of core concepts discussed in this course, including op-amps, integrators, differentiators, cantilever beams, and strain gauges.

Theory:

As the aim of this report is to estimate the velocity of a cantilever beam, kinematics is a core concept that must be understood throughout the project. Velocity is defined as the rate of change of displacement over time, and acceleration is the rate of change of velocity over time. The fundamental relationship between displacement ($x(t)$), velocity ($v(t)$), and acceleration ($a(t)$) is as follows:

$$\begin{aligned}\text{Velocity (m/s): } v(t) &= x'(t) = \frac{dx}{dt} = \frac{\Delta x}{\Delta t} \\ \text{Acceleration (m/s}^2\text{): } a(t) &= v'(t) = \frac{d^2x}{dt^2} = \frac{dv}{dt} = \frac{\Delta v}{\Delta t}\end{aligned}$$

By using the relationships as shown above, velocity can be calculated by integrating the acceleration measurement obtained using an accelerometer. A Miller Integrator was used to perform this integration. To measure the acceleration using the accelerometer device, capacitor plates are placed around the internal mass of the accelerometer. Therefore when the accelerometer experiences motion, the mass shifts, causing changes in capacitance. These variations in capacitance modify the sensor's output, with the amplitude of the signal being proportional to the acceleration.

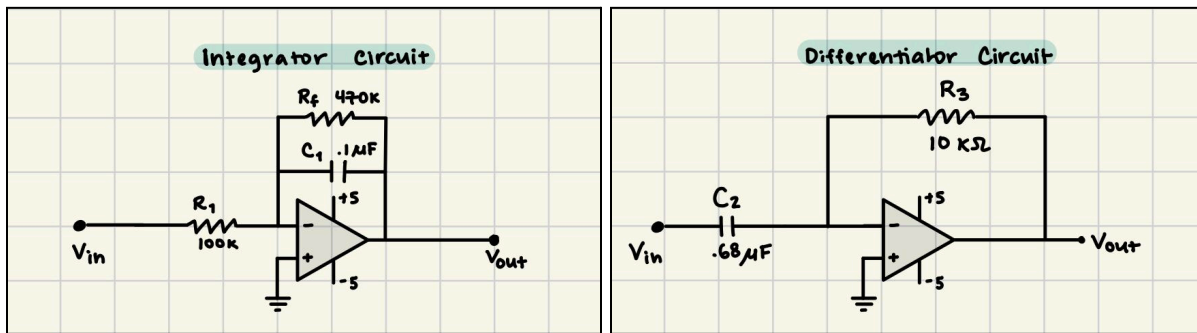
The relationship between displacement and velocity will also be greatly beneficial in measuring the terminal velocity of a cantilever beam and a strain gauge can be utilized to do so. A strain gauge works by detecting changes in electrical resistance when the object it is attached to undergoes axial, bending, shear, or torsional forces. Since electrical resistance changes in proportion to the strain experienced by the material, the strain gauge can be used to quantify deformation. The resulting change in resistance produces a signal that is proportional to the displacement of the beam's end. As the derivative of displacement measures the velocity, a differentiator is used to perform this derivation.

The strain gauge circuit usually consists of 2 basic elements: a Wheatstone Bridge and a differential amplifier. The Wheatstone Bridge is a circuit consisting of 4 resistances with a voltage connection. This bridge is mainly used to measure the small changes in the resistance of

the strain gauge and convert it into a measurable voltage signal. When in an imbalance state due to strain, the Wheatstone Bridge will output a small differential voltage that is proportional to strain. After this signal output, a differential amplifier is used to amplify this differential voltage while reducing environmental factors to increase measurement accuracy.

For this project, both the integrator and a differentiator circuit will be used. The Miller Integrator, which is used to perform the integration of acceleration measurement, consists of an operational amplifier (op-amp) with a feedback capacitor. In addition, it is expected for the integrator to act as an integrator at frequencies where the input signal frequency is above the low-frequency cutoff of ~ 21.3 radians/s ($\omega_{cutoff} = \frac{1}{R_f C} = \frac{1}{(470k)(0.1\mu F)}$).

The differential circuit consists of an op-amp with a capacitor at the input and a resistor in feedback and is used to differentiate the displacement outputted by the strain gauge to estimate velocity. The differentiator functions as a differentiator at frequencies below the cutoff frequency of ~ 147 radians/s ($\omega_{cutoff} = \frac{1}{R_f C} = \frac{1}{(10k)(.68\mu F)}$). The component values for both the integrator and the differentiator are shown in the figure below.



Figures 1 & 2: Integrator and Differentiator Circuits Sketch

The capacitance and resistance values that make up the miller integrator circuit are reasonable as they allow for the smooth operation of the circuit. The product between the resistance and the capacitance is equivalent to the time constant, which determines how quickly the output ramps for a given input. The integrator RC value must be large enough to perform effective integration and the differentiator RC must be small enough. This value directly influences the charging and discharging qualities of the capacitor. In the integrator circuit, the calculated time constant using the component values as shown in the figure above is 47 ms. These values are reasonable as RC is greater than the inverse of the beam's oscillating frequency. In the differentiator circuit, the RC value was calculated to be 6.8 ms, which is appropriately small for effective differentiation without excessive noise amplification.

For this project, an OP27 with a voltage of $\pm 5V$ was used, meaning the output will saturate at an amplitude of 5V. To avoid saturation for these op-amps, the gain equation can be used to

determine the maximum input voltage for the integrator and the differentiator. The gain of the Miller Integrator is given by $\frac{V_{out}}{V_{in}} = -\frac{1}{j\omega R_{in} C_f}$ and the gain of the differentiator is given by $\frac{V_{out}}{V_{in}} = -j\omega R_f C_{in}$. From these equations, the maximum input signals for the integrator and the differentiator are ~5V and ~9V, respectively.

Calibration:

A calibration process must be utilized to generate usable data for the experiments. The following equation to calibrate the accelerometer output signal to m/s² units was given by the experiment manual:

$$a_b[t] = (V_a[t] - 1.5) \frac{9.8}{0.420}$$

The ADXL327 accelerometer datasheet specifies that the output sensitivity is 420 mV/g, where g represents the acceleration due to gravity (9.8 m/s²). At zero acceleration, the output voltage is centered at 1.5V, which is midway between ground and the 3V power supply voltage.

For this project, the strain gauge must be calibrated as well by measuring the output voltage as a function of the beam's displacement. With 5 displacement values selected (-1 cm, -0.5 cm, 0 cm, 0.5 cm, and 1 cm) along with corresponding voltages, V_{out} vs. Position can be generated using Excel to determine the sensitivity, k_1 . The following equation was used to determine the calibrated change in position relative to the output voltage:

$$x_b[t] = (V_{sg}[t]) \frac{1}{k_1}$$

Gain Constraints:

The gain constant in the interior circuit is found by solving the transfer function. The expected gain for the integrator with the given components is ~0.6 and the expected gain for the differentiator with the given components is ~0.9.

Circuit Operation:

Accelerator Circuit

The accelerator circuit detects acceleration and converts it into an electrical signal, which is then processed by the integrator. The integrator itself is made of an amplifier with a capacitor; this converts the acceleration signal into velocity. Figure 3 shows the hand drawing of the accelerator circuit.

The output of the integrator was aligned with the measured acceleration. This would be noted by the displayed amplitude and phase pattern. If properly aligned, the input is detected. Despite this,

there may be slight deviations in the data, considering the conduction of resistors, tampering with components of the circuit, and non-ideal behavior in the amplifier.

Refer to Appendix C for velocity plots.

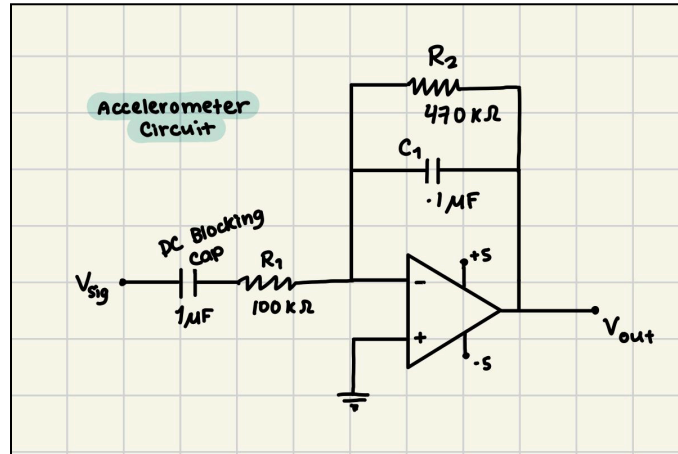


Figure 3: Accelerometer Circuits Diagram

Strain Gauge Circuit

Yes, the output of the differentiator was fairly consistent with the measured strain. The phase shift would be 90 degrees and the acceleration amplitude would be 0.8 m/s^2 . This is because the acceleration plot shows values oscillating from 34.2 m/s^2 to 35.8 m/s^2 .

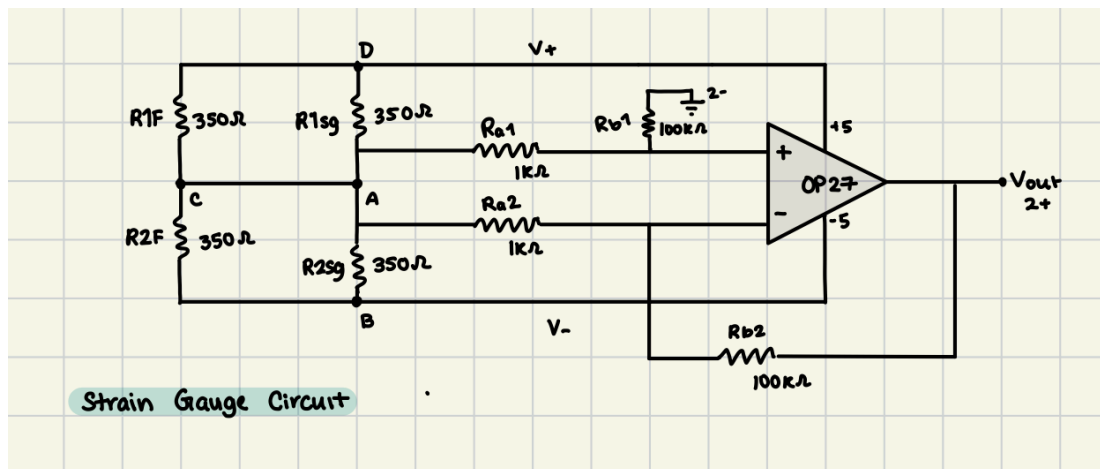


Figure 4: Strain Gauge Circuit Diagram

Final Analysis and Conclusion

What was the peak acceleration of the beam?

- The peak acceleration of the beam differed between tests but it was around 1.8 m/s^2

What was the peak deflection?

- Around .42m

Are these traces consistent if you assume the motion is a pure sine wave? Amplitude and phase? If not, speculate why not.

- The traces deviate from a pure sine wave due to periodic and random spikes in amplitude. However, the phase remains relatively consistent, maintaining a shape that resembles a true sine wave.

Compare strain gauge and accelerometer measurement of velocity. Is one better suited to determine the velocity? If so, why? What is the peak velocity measured? Do the calculated velocities have the same amplitudes and phases?

- The accelerometer proved to be significantly more effective for measuring velocity. When generating graphs from our data, those based on the accelerometer readings were noticeably more consistent and exhibited fewer errors compared to those from the strain gauge. Overall, the accelerometer demonstrated greater precision in determining velocity.
- The peak velocities recorded were 0.36 m/s using the accelerometer and 0.7 m/s using the strain gauge. This discrepancy is notable, likely due to the strain gauge exhibiting a much wider variation and more pronounced spikes in its data.

How could each measurement be improved? What are the errors associated with each measurement? State your conclusions on measurements of strain, acceleration, and velocity, as they apply to the instrumented beam.

- Measurement accuracy could be improved by minimizing sources of human error. The strain gauge is highly sensitive to small tensions and compressions, so even a slight accidental bump or contact could significantly affect the readings. Similarly, improper handling of the accelerometer can introduce vibrations in the beam, resulting in greater variability in the data.
- Improper results could also stem from loose wires or an incorrect setup, further emphasizing the importance of minimizing human error to improve the experiment's accuracy. As their names suggest, strain was measured using the strain gauge and acceleration with the accelerometer. While both devices were used to determine velocity, the accelerometer consistently produced more reliable and stable data.

Accelerometers are used extensively these days in cars. How would you use accelerometer signals in a car to enhance the driving experience? If there are so many accelerometers in present-day cars, why is acceleration typically not displayed for the driver?

- One common use of an accelerometer in a car is in the deployment of airbags. When it detects a sudden deceleration (such as from a collision) it immediately sends a signal to activate the airbags, helping protect the passengers. In this context, the accelerometer must operate with exceptional speed and reliability, as it's critical to vehicle safety across the entire automotive industry.
- Unlike speed, acceleration is not a regulated metric for drivers and can often be difficult to interpret in real-time. For this reason, it's generally not displayed on a car's dashboard, as it could cause confusion rather than provide useful information.

If you had a portable accelerometer, what would you do with it?

- If we had access to a portable accelerometer, we could explore its applications in both vehicles and buildings. In cars, we could investigate how advancements in accelerometer technology might enhance overall safety systems. For buildings, a portable accelerometer could be used to monitor structural vibrations, detect seismic activity, and potentially serve as an early warning tool for earthquakes.

Personal Efforts:

Larissa: 25% - Aided M2k circuitry, read the entire lab beforehand, attended open shop hours to complete the experiment

Emily: 25% - M2k circuitry, interpreted Scopy, attended open shop hours to complete experiment

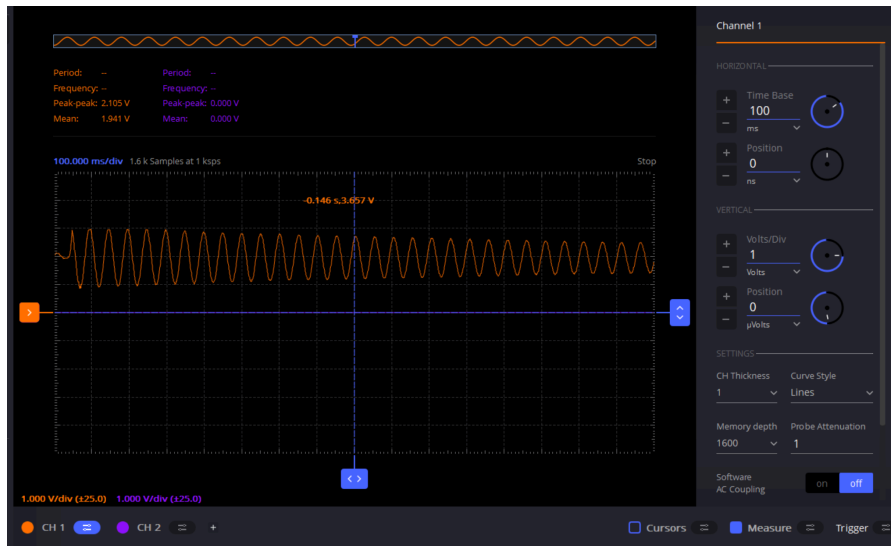
Pooja: 25% - Hand-drawn circuit diagrams, Matlab plots, write-ups: calibration process, gain constraints

Rachael: 25% - Aided M2k circuitry, read entire lab beforehand, attended open shop hours to complete experiment, write-ups: background & theory

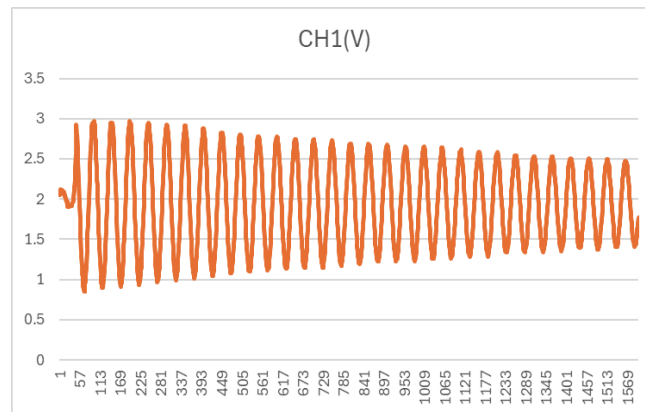
Appendix

Appendix A: Accelerometer

1. Plot of raw acceleration data vs. time, signed.

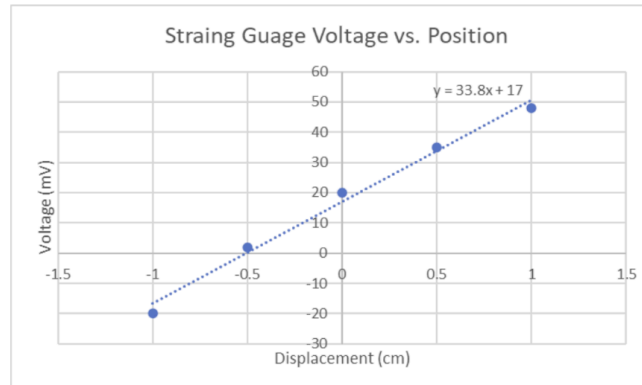


2. Plot of acceleration including calibration constant.



Appendix B: Strain and Accelerometer

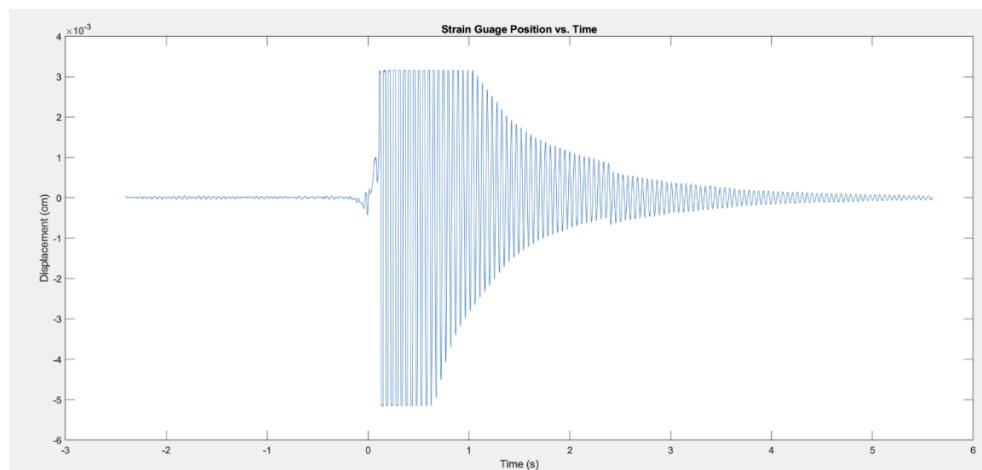
1. Plot of strain gauge voltage vs. position. Indicate the slope of data.



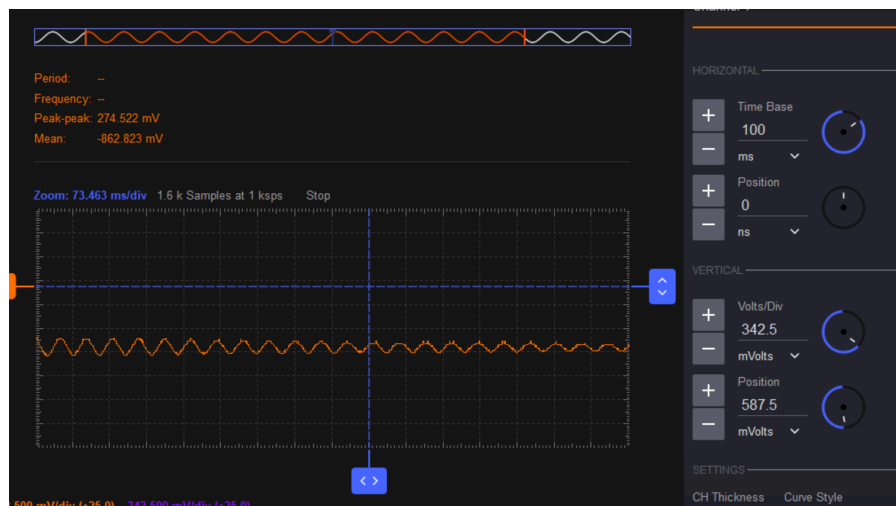
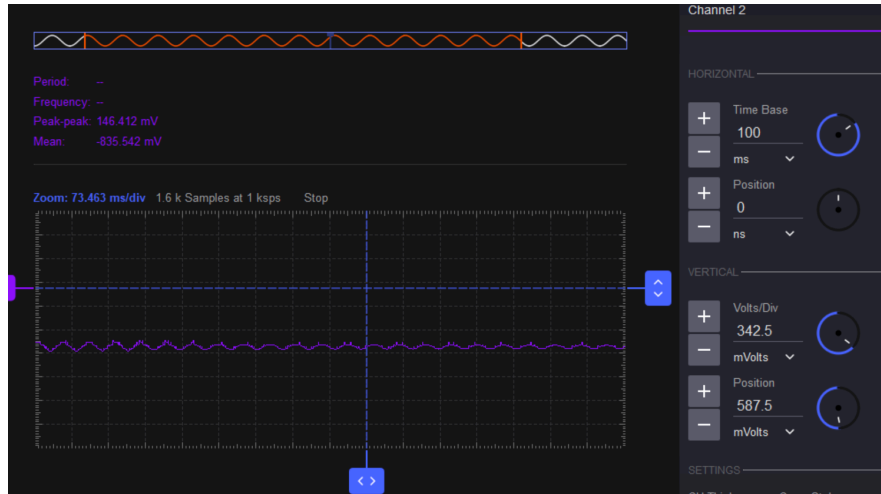
2. Plot of raw position data vs. time signed.



3. Plot of position vs. time, includes the calibration constant.

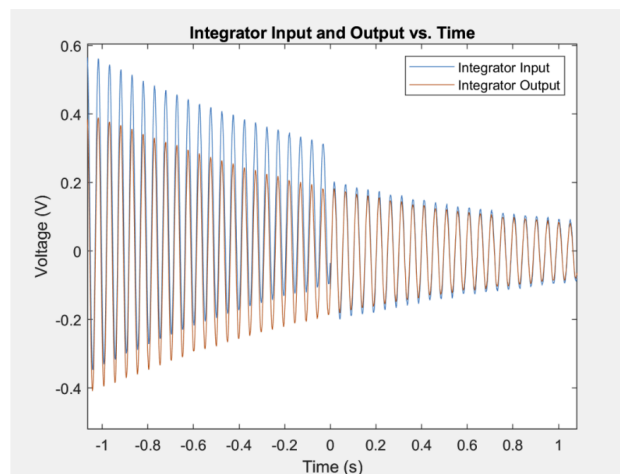


4. Plot of raw from accelerometer and strain gauge, signed.

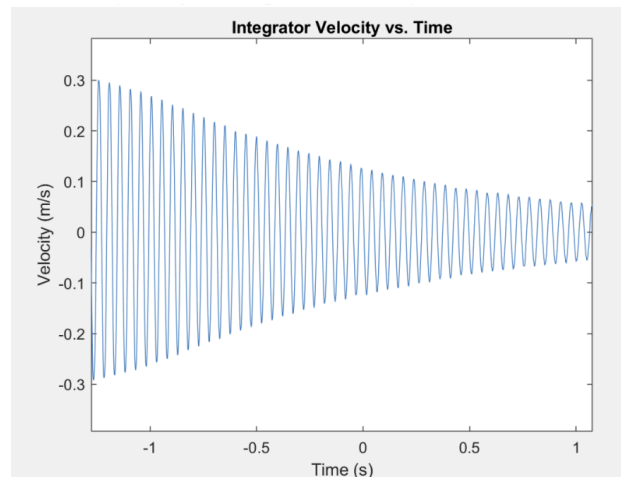


Appendix C: Velocity

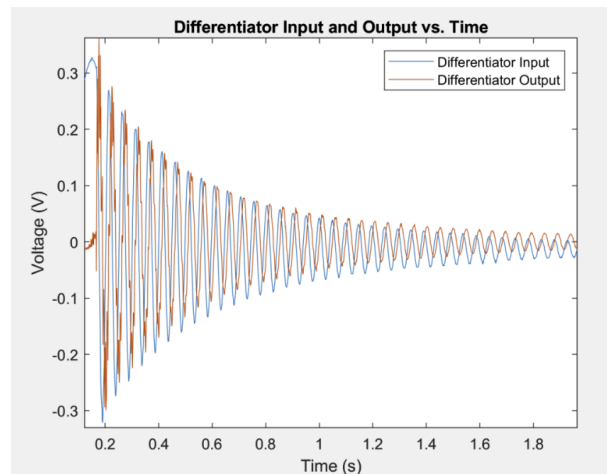
1. Plot of input and output of the integrator signed.



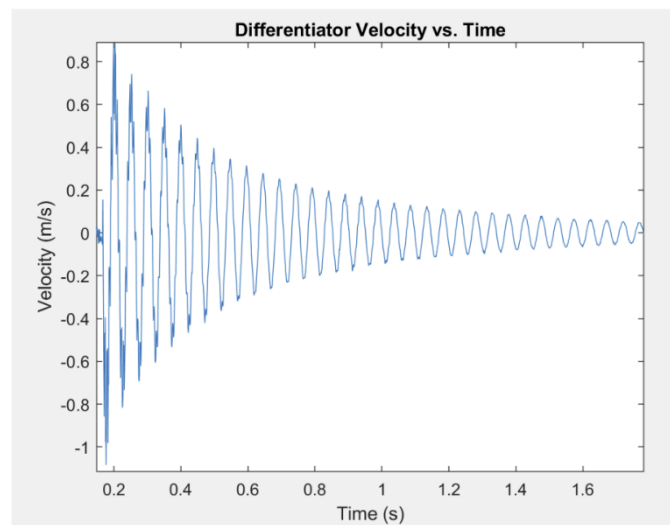
2. Plot of velocity using the output of the integrator, scaled and labeled.



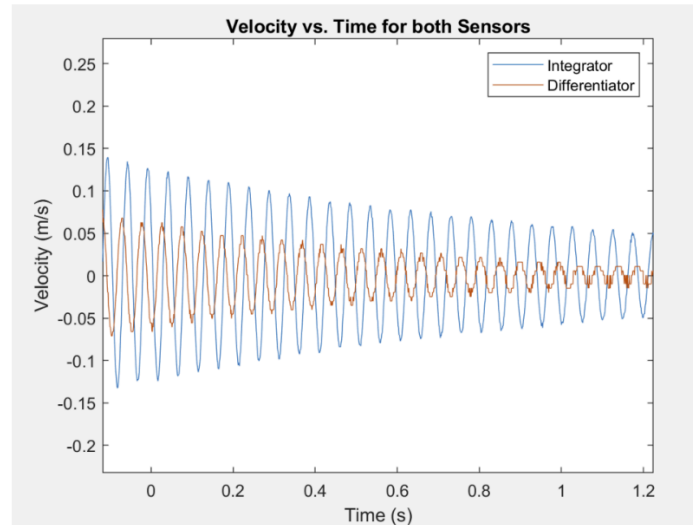
3. Plot of input and output of the differentiator signed.



4. Plot of velocity using the output of the differentiator, scaled and labeled.



- Plot of velocity from sensors on the same plunk, scaled and labeled.



Appendix D: References (Must be included.)

- <https://www.dwyeromega.com/en-us/resources/accelerometers#:~:text=As%20the%20sensor%20experiences%20acceleration,of%20acceleration%20on%20the%20sensor.>
- <https://www.ni.com/en/shop/data-acquisition/sensor-fundamentals/measuring-strain-with-strain-gages.html#:~:text=A%20strain%20gage%20works%20by,be%20used%20to%20quantify%20strain>
- <https://www.allaboutcircuits.com/textbook/semiconductors/chpt-8/differentiator-integrator-circuits/>
- [https://eng.libretexts.org/Bookshelves/Electrical_Engineering/Electronics/Operational_Amplifiers_and_Linear_Integrated_Circuits_-_Theory_and_Application_\(Fiore\)/10%3A_Integrators_and_Differentiators/10.02%3A_Integrators#:~:text=This%20is%20a%20very%20important,Response%20of%20a%20practical%20integrator.&text=Determine%20the%20Equation%20for%20V,6.&text=The%20lower%20limit%20of%20integration%20is%20set%20by%20flow.&text=This%20represents%20our%2050%25%20accuracy,to%200higher%20and%20higher%20frequencies.](https://eng.libretexts.org/Bookshelves/Electrical_Engineering/Electronics/Operational_Amplifiers_and_Linear_Integrated_Circuits_-_Theory_and_Application_(Fiore)/10%3A_Integrators_and_Differentiators/10.02%3A_Integrators#:~:text=This%20is%20a%20very%20important,Response%20of%20a%20practical%20integrator.&text=Determine%20the%20Equation%20for%20V,6.&text=The%20lower%20limit%20of%20integration%20is%20set%20by%20flow.&text=This%20represents%20our%2050%25%20accuracy,to%200higher%20and%20higher%20frequencies.)
- [https://www.keysight.com/used/us/en/knowledge/formulas/time-constant-formula#:~:text=Identify%20the%20resistance%20\(R\)%20and,L/R%20for%20RL%20circuits.](https://www.keysight.com/used/us/en/knowledge/formulas/time-constant-formula#:~:text=Identify%20the%20resistance%20(R)%20and,L/R%20for%20RL%20circuits.)

Appendix E: Extra Credit Plots

N/A